Health watch — main board

Sơ đồ phân cấp: mỗi ô là một trang con. Nhãn toàn cục (global label) nối tín hiệu giữa các trang; ký hiệu nguồn nối theo tên.

Ref 101–199. Sinh tự động bởi tools/gen_sch.py từ tools/design.py.

Power: USB-C, AXP2101 PMIC, battery, current shunts

File: health_watch_main_power.kicad_sch

ESP32-S3R8 + 16 MB flash + RF

File: health_watch_main_mcu.kicad_sch

Display FPC, IMU, RTC, LED boost, sensor-board FPC

File: health_watch_main_periph.kicad_sch

Motor, buzzer, SOS button, status LED

File: health_watch_main_ui.kicad_sch

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harisama1 — đồ án đồng hồ sức khỏe

Sheet: /
File: health_watch_main.kicad_sch

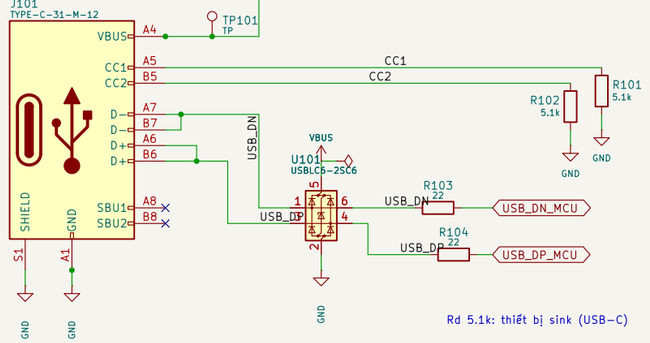
Title: Health watch — main board — root

Size: A4	Date: 2026-11-05	**Rev: 1.2**
KiCad E.D.A. 10.0.8		Id: 1/5

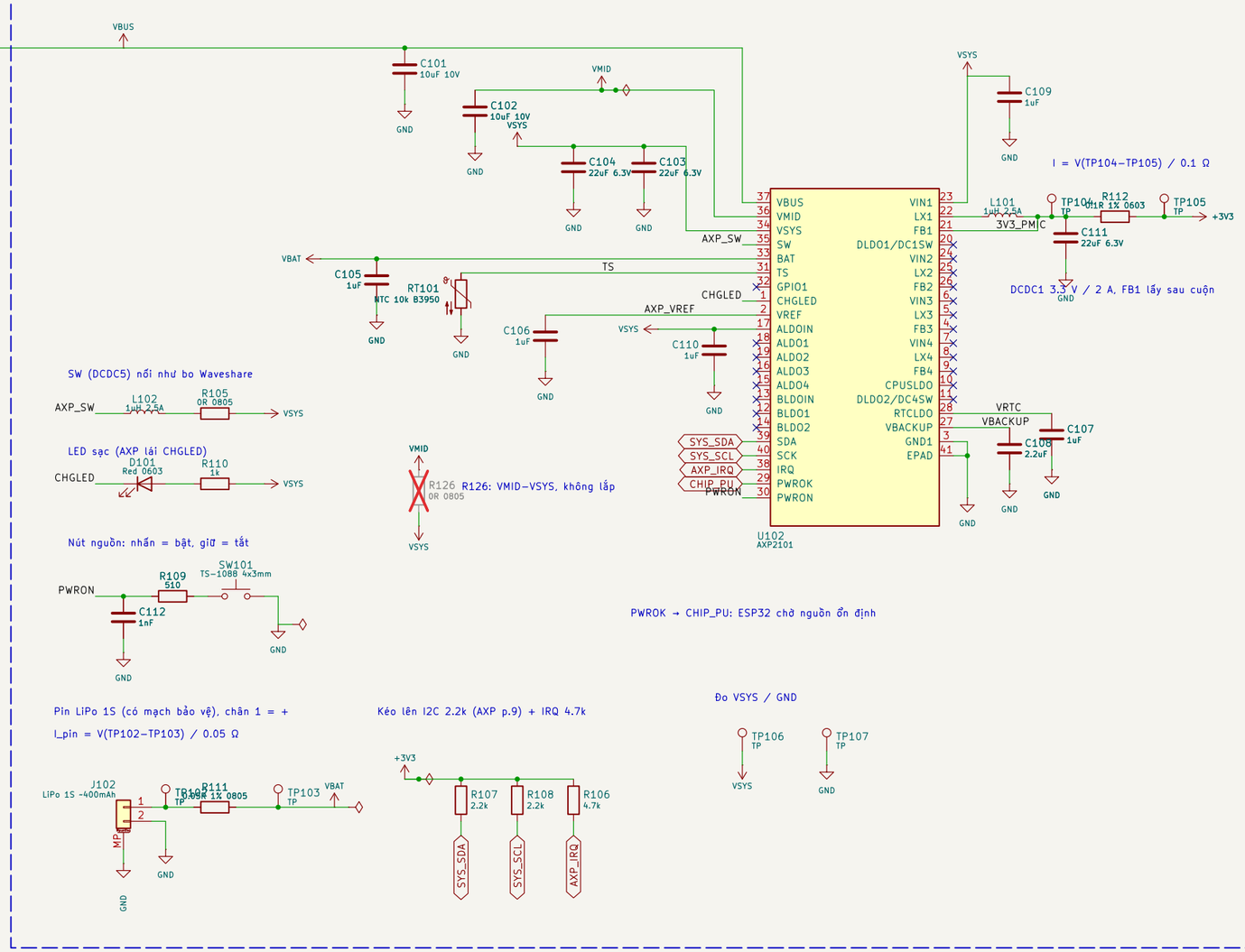
Power: USB-C, AXP2101 PMIC, battery, current shunts

USB-C → ESD → AXP2101 (sạc LiPo + DCDC1 3.3 V). Shunt + test point để đo dòng: pin (R111) và 3V3 (R112).

USB-C + ESD



PMIC AXP2101 + pin + shunt đo dòng



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harisama1 — đồ án đồng hồ sức khỏe
Sheet: /Power: USB-C, AXP2101 PMIC, battery, current shunts/
File: health_watch_main_power.kicad_sch
Title: Health watch — main board — Power: USB-C, AXP2101 PMIC, battery,
Size: A3 | Date: 2026-11-05 | **Rev: 1.2**
KiCad E.D.A. 10.0.8 | Id: 2/5

ESP32-S3R8 + 16 MB flash + RF

ESP32-S3R8: decoupling theo HDG Fig.1 p.5, thạch anh 40 MHz, flash QSPI 16 MB, mạch phối hợp RF + ăng-ten gốm, UART0.

ESP32-S3R8 + flash + thạch anh + RF + UART0

VDD_SPI = 3.3 V (GPIO45 = 0 lúc reset) cấp flash + PSRAM

VDD3P3 (PA RF): lọc LC 2.0 nH

+3V3
L104 2.0nH
VDD3P3
C115 10uF 16V
C116 1uF
C117 100nF
GND
C118 1uF
C119 10nF
C120 100nF
C121 100nF
C122 100nF
C123 1uF
VDD_SPI

U104 ESP32-S3R8

VDDA	55		
VDD3P3	12		
VDD3P3_RTC	20		
VDD_SPI	29		
VDD3P3_CPU	46		
GND	57		
BOOT	5	GPIO0	
LCD_RST	6	GPIO1	
TP_RST	7	GPIO2	
VLED_EN	8	GPIO3	
QSPI_D0	9	GPIO4	
QSPI_D1	10	GPIO5	
QSPI_D2	11	GPIO6	
QSPI_D3	12	GPIO7	
BTN_SOS	13	GPIO8	
IMU_INT1	14	GPIO9	
PPG_INT	15	GPIO10	
TP_INT	16	GPIO11	
LCD_CS	17	GPIO12	
LCD_TE	18	GPIO13	
SYS_SCL	19	GPIO14	
RTC_INT	23	GPIO17	
LED_STAT	24	GPIO18	
USB_DN_MCU	25	GPIO19/USB_D-	
USB_DP_MCU	26	GPIO20/USB_D+	
IMU_INT2	27	GPIO21	
SYS_SDA	21	XTAL_32K_P	
AXP_IRQ	22	XTAL_32K_N	
	54	XTAL_P	
	53	XTAL_N	
	28	SPICS1	
SPIHD	30	SPIHD	SPIHD
SPIWP	31	SPIWP	SPIWP
SPICS0	32	SPICS0	SPICS0
SPICLK	33	SPICLK	SPICLK
SPIQ	34	SPIQ	SPIQ
SPID	35	SPID	SPID
SPICLK_N	36		
SPICLK_P	37		
GPIO33	38		
GPIO34	39		
GPIO35	40		
GPIO36	41		
GPIO37	42		
GPIO38	43	QSPI_CLK	
MTCK	44	PPG_SCLK	
MTDO	45	PPG_MOSI	
MTDI	47	PPG_MISO	
MTMS	48	PPG_CS	
U0TXD	49	U0TXD_MCU	
U0RXD	50	U0RXD_MCU	
GPIO45	51	BUZZER	
GPIO46	52	MOTOR	
CHIP_PU	4	CHIP_PU	
LNA_IN	1		

J103 UART0 1x03 2.54
R118 499
L105 24nH
Y101 40MHz 10ppm CL15pF
C125 22pF C0G
C126 22pF C0G
40 MHz, CL 15 pF ≈ 22·22 / 44 + 4 pF

L106 2.4nH
RF1
R117 0R
RF_ANT
AE101 2450AT18A100E
C128 1.5pF C0G
C129 1.5pF C0G
C130 DNP
C131 DNP
Phối hợp phía chip (HDG Table 6) | mạch π phía ăng-ten (DNP, tune bằng VNA)

U105 W25Q128JVSIQ

SPICS0	1	CS	
SPICLK	6	CLK	
SPID	5	DI/IO0	
SPIQ	2	DO/IO1	
SPIWP	3	WP/IO2	
SPIHD	7	HOLD/RESET/IO3	
VCC	8		
GND	4		

C127 100nF

Reset (CHIP_PU) + Boot (GPIO0)

R116 10k
SW103 TS-1088 4x3mm
Giữ BOOT + nhấn RESET = chế độ nạp

R115 10k
C124 1uF
SW102 TS-1088 4x3mm
RC 10k/1µF: CHIP_PU lên sau nguồn ≥ 50 µs (HDG §1.3.3)

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Sheet: /ESP32-S3R8 + 16 MB flash + RF/
File: health_watch_main_mcu.kicad_sch

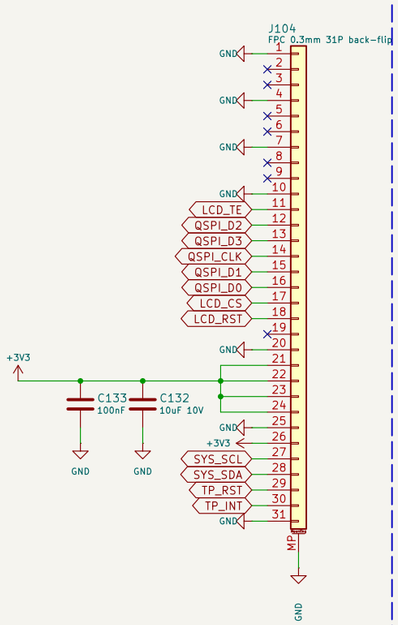
Title: Health watch — main board — ESP32-S3R8 + 16 MB flash + RF

Size: A3	Date: 2026-11-05	**Rev: 1.2**
KiCad E.D.A. 10.0.8		Id: 3/5

Display FPC, IMU, RTC, LED boost, sensor-board FPC

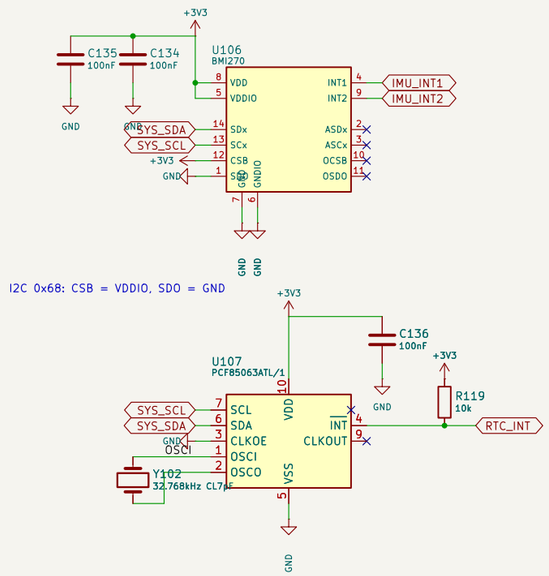
Màn hình AMOLED (FPC 31P như Waveshare 1.75C), IMU BMI270, RTC PCF85063A, boost 5 V cho LED PPG, FPC 16P sang bo cảm biến.

FPC màn hình AMOLED + cảm ứng



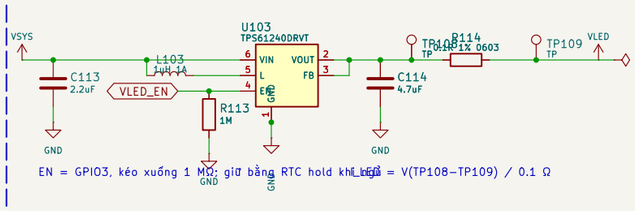
QSPI + I2C chung

IMU BMI270 + RTC PCF85063A



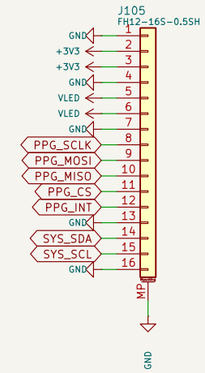
I2C 0x68: CSB = VDDIO, SDO = GND

Boost 5 V cho LED PPG (TPS61240)



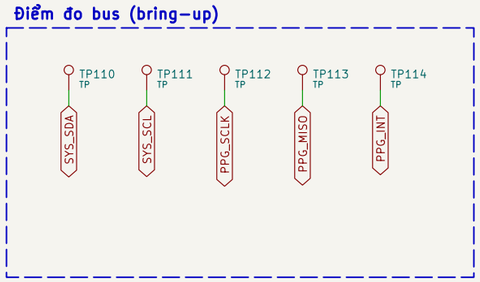
EN = GPIO3, kéo xuống 1 MΩ: giữ bằng RTC hold khi ngủ. I_LED = V(TP108−TP109) / 0.1 Ω

FPC 16P sang bo cảm biến



Cáp đảo đưa 5 V vào chân tín hiệu: đo thông mạch trước khi cấp nguồn

Điểm đo bus (bring-up)



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harisama1 — đồ án đồng hồ sức khỏe

Sheet: /Display FPC, IMU, RTC, LED boost, sensor-board FPC/
File: health_watch_main_periph.kicad_sch

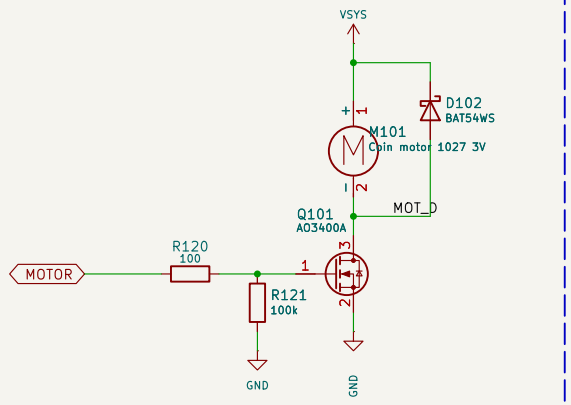
Title: Health watch — main board — Display FPC, IMU, RTC, LED boost, senso

Size: A3	Date: 2026-11-05	**Rev: 1.2**
KiCad E.D.A. 10.0.8		Id: 4/5

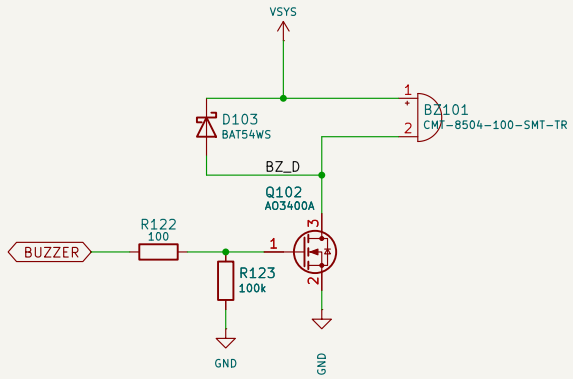
Motor, buzzer, SOS button, status LED

Motor rung + còi từ: MOSFET phía thấp, diode dập, cổng kéo xuống (GPIO45/46 là chân strapping). Nút SOS, LED trạng thái.

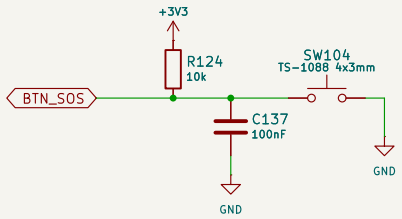
Motor rung (GPIO46)



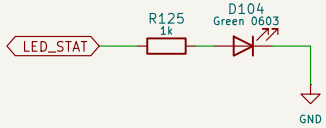
Còi từ (GPIO45, PWM ~2.7 kHz)



Nút SOS (GPIO8, đánh thức deep-sleep)



LED trạng thái (GPIO18)



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harisama1 — đồ án đồng hồ sức khỏe

Sheet: /Motor, buzzer, SOS button, status LED/
File: health_watch_main_ui.kicad_sch

Title: Health watch — main board — Motor, buzzer, SOS button, status LED

Size: A4	Date: 2026-11-05	**Rev: 1.2**
KiCad E.D.A. 10.0.8		Id: 5/5